

SHEKINAH OKUNBO

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Electrical and Computer Engineering, Data Science Minor

Expected Graduation: May 2029

- **Relevant Coursework:** Fundamentals of Computer Engineering (computer architecture, digital logic, embedded C), Data Structures and Algorithms, Object-Oriented Programming, Linear Algebra, Signals and Systems (in progress), NVIDIA Autonomous Robotics
- **Honors and Programs:** NVIDIA Summer Bridge (2026); Rice Datathon 2026, 1st Place (Underclassmen Division); NMC National Physics Olympiad 2025, 4th Nationally; Jane Street FOCUS (2026)

TECHNICAL SKILLS

Languages: C++ (C++17), C, CUDA, Python, TypeScript, JavaScript, SQL

Systems and Tools: Linux, Docker, Kubernetes, Git, GitHub Actions (CI/CD), ThreadSanitizer, AddressSanitizer, FastAPI, AWS (Lambda, SQS, DynamoDB), Code Composer Studio

Machine Learning: PyTorch, ONNX, YOLO, model quantization, on-device inference, NumPy

Concepts: GPU programming, parallel programming, multithreading, lock-free data structures, computer architecture (pipelining, caches), embedded systems, data structures and algorithms

EXPERIENCE

Rice University, Department of Electrical and Computer Engineering

Houston, TX

Machine Learning Researcher, Summer Undergraduate Research Fellowship

May 2026 – Jul 2026

- Identified the fastest deployable vision model at 30.08 FPS and 15.25 ms latency, a 3.2x latency cut over the largest variant, by benchmarking five YOLOv2 variants in PyTorch on an iPhone 16 Pro.
- Diagnosed quantization and format compatibility, not model architecture, as the constraint blocking GPU-scale models from running on-device by profiling the export and deployment pipeline.
- Exposed a misleading accuracy result, 100% recall at 4.7% precision from a 53-image training set, by training custom detectors and showing that dataset scale governed reliability; presented the findings as a research poster.

RiceApps

Houston, TX

Software Developer, Adaptive Learning Platform

Sep 2025 – Present

- Shipped a production GED exam prep platform serving 500+ students for the nonprofit Eight Million Stories by designing a normalized PostgreSQL schema and SQL migrations behind a diagnostic quiz that routes each student by weak area.
- Delivered 197+ peer-reviewed commits to a TypeScript and Next.js codebase inside a 70-person student engineering organization.

PROJECTS

WarpServe: GPU-Accelerated Vision Preprocessing Service | C++, CUDA, Python, Docker, Kubernetes

Aug 2026

- Accelerated image preprocessing 127x over a scalar CPU reference on an NVIDIA T4 GPU, with bit-identical output, by hand-writing a CUDA kernel that fuses bilinear resize, normalization, and HWC-to-CHW layout conversion into one pass.
- Measured the gain at only 1.7x once host-to-device and device-to-host copies were included, isolating PCIe data transfer, not GPU compute, as the bottleneck that should drive the serving architecture.
- Doubled GPU matrix-multiply throughput (1.97x, from 411 to 812 GFLOP/s) over a naive kernel by tiling through shared memory, a pure memory-locality gain with identical arithmetic.
- Scaled the Dockerized service (CPU build, local Kubernetes cluster) from 2 to 8 pods to 88 requests/second at 83% scaling efficiency after diagnosing that kubectrl port-forward, not the pods, capped throughput.

EdgePulse: Real-Time Forecasting and Anomaly Detection | C++17, Python, multithreading, ctypes

Dec 2025 – Oct 2026

- Raised peak throughput 2.2x, from 85K to 189K messages/second, and cut median latency 3.2x by replacing a mutex-guarded queue with a lock-free single-producer single-consumer ring buffer between C++ ingest and detection threads.
- Reduced forecasting latency 31% versus pure Python, reaching 508K predictions/second, by moving the hot path into a C++ core called through ctypes on live Coinbase price feeds.
- Verified thread and memory safety by running unit tests under ThreadSanitizer and AddressSanitizer in GitHub Actions, with an allocation-free C++ detector that matched the Python pipeline with 0 alert mismatches over 20,000 ticks.

TI MSPM0 Gamepad Controller (course project) | C, ARM Cortex-M0+, Code Composer Studio

Jan 2026 – Apr 2026

- Improved input responsiveness on a bare-metal gamepad by moving analog joystick sampling through the ADC from continuous polling to timer-driven sampling, then to interrupt service routines.
- Controlled GPIO and ADC peripherals without a hardware abstraction layer by mapping C structs and pointers onto device registers, and debugged by inspecting the program counter, stack, and SRAM in the hardware debugger.

LearnStream: Event-Driven Analytics Pipeline | Python, AWS Lambda, SQS, DynamoDB, CDK

Sep 2026

- Achieved zero failures across 6,898 requests at 38 events/second and 160 ms p99 latency by tracing intermittent errors to Lambda concurrency limits; verified exactly-once processing by replaying one event 25 times to store exactly 1 record.

ACTIVITIES

ColorStack Fellow | NSBE (National Society of Black Engineers) | Rice IEEE | CodePath | HackRice